CO oxidation on IrO₂(110) surfaces

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ABSTRACT

We investigated the oxidation of CO on stoichiometric and O-rich IrO₂(110) surfaces using temperature programmed reaction spectroscopy (TPRS), density functional theory (DFT) calculations and microkinetic simulations. Adsorbed CO on the s-IrO₂(110) surface generates CO and CO₂ peaks near 545 K during TPRS, and only about 38 % of the CO oxidized to CO₂ when the initial CO layer was saturated. Pre-adsorbed O-atoms, so-called on-top oxygen (O_t), promote the oxidation of CO adsorbed on IrO₂(110). On the O_t-covered surface, CO oxidation by O_t atoms produces a CO₂ TPRS peak at ~370 K, and all of the initially adsorbed CO oxidizes to CO₂ when the initial O_t coverage is greater than the CO coverage. In agreement with the TPRS results, DFT calculations predict that the barrier is about 100 kJ/mol lower for CO oxidation by an O_t atom than a lattice O-atom of IrO₂(110). A microkinetic model, parameterized with energy barriers computed using DFT, accurately reproduces the CO and CO₂ TPRS traces only after CO binding energies are lowered to values determined using a hybrid exchange-correlation functional and the barrier for CO molecules to fill bridging O-vacancies is lowered. The simulations predict that O-vacancies play an important role in mediating the CO oxidation kinetics on s-IrO₂(110), and thereby demonstrate the importance of future spectroscopic studies aimed at characterizing the nature of the surface CO and O species involved in reaction. This study provides new insights for understanding CO oxidation on IrO₂(110), and provides evidence that several elementary steps can be involved in governing this chemistry.

1. Introduction

Materials that incorporate IrO₂ have potential to serve as efficient catalysts for the complete or partial oxidation of alkanes, and are utilized in applications of electrocatalysis as well. Prior studies show that the rutile IrO₂(110) facet promotes the activation of light alkanes (C1–C3) at temperatures as low as 100 K, and that the resulting hydrocarbon fragments react to evolve oxidation products above about 400 K during temperature programmed reaction spectroscopy (TPRS) [1–6]. Rutile IrO₂ is also highly active in promoting the oxidation of CH₄ and C₂H₆ under catalytic conditions, exhibiting high selectivity toward complete oxidation products (CO₂, H₂O) [7–10]. Computational investigations predict that the oxidation of CO is an essential step in the mechanism governing complete alkane oxidation on IrO₂(110) [4,7,11]. As such, understanding the interactions of CO with the IrO₂(110) surface can help to inform the development of IrO₂-based materials for catalytic

hydrocarbon oxidation. Moreover, the oxidation of CO has served as a prototype reaction for characterizing elementary reaction processes on metal and metal-oxide surfaces, and is an important reaction in catalytic applications of exhaust gas remediation as well.

The interactions of CO on the RuO₂(110) and PdO(101) surfaces have been extensively investigated, with such studies originally inspired by observations that metal-oxides can develop as thin layers under oxygen-rich conditions and can be highly active toward catalytic CO oxidation [12–19]. Given their structural and chemical similarities, it is reasonable to expect that CO oxidation on IrO₂(110) will involve similar steps as reported for RuO₂(110) and PdO(101). While surface science investigations of CO oxidation on IrO₂(110) have not yet been reported, a recent study shows that rutile IrO₂ is active for catalytic CO oxidation, but exhibits lower activity than RuO₂ [20]. The present study provides insights for understanding the lower CO oxidation activity of IrO₂ compared with RuO₂.

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Characterizing the reactivity of chemisorbed O-atoms vs. lattice O-atoms toward the oxidation of CO on IrO₂(110), i.e., on-top vs. bridging O-atoms (O_t vs. O_{br}), is particularly important for understanding CO oxidation under catalytic conditions since O_t atoms are typically generated by O₂ in the reactant feed. We have previously shown that O_t atoms promote the complete oxidation of H₂, CH₄ and C₂H₆ on IrO₂(110) by providing low energy pathways for H₂O desorption, as evidenced by the emergence of lower temperature H₂O TPRS peaks when O_t atoms are adsorbed prior to H₂ or alkane adsorption [4,5,7,21,22]. Compared with O_{br} atoms, the removal of O_t atoms during H₂O formation is a lower barrier process because the O_t atoms bind more weakly on the surface. Consistent with these findings, a recent computational study predicts that CO oxidation by O_t atoms is also more favorable than reaction with O_{br} atoms on IrO₂(110) [23]. Interestingly, however, prior studies suggest that O_t and O_{br} atoms exhibit similar activity toward CO oxidation on RuO₂(110), even though O_t atoms bind more weakly on this surface as well [12,13,15,24–26]. Characterizing the reactivity of O_t and O_{br} atoms toward CO oxidation on IrO₂(110) is important for understanding catalytic applications, and can provide an interesting comparison with prior results obtained with RuO₂(110).

In the present study we investigated CO oxidation on stoichiometric (s-) and O_t-rich IrO₂(110) surfaces using TPRS, DFT calculations and microkinetic simulations. Our results show that CO oxidation on s-IrO₂(110) produces CO₂ in a TPRS peak at about 545 K, and that only about 38 % of the adsorbed CO oxidizes to CO₂ when the initial CO layer is saturated. We find that O_t atoms are significantly more active than O_{br} atoms on IrO₂(110) for promoting CO oxidation during TPRS as they provide a lower energy pathway for reaction and enhance the probability for CO oxidation over desorption. Our DFT calculations and microkinetic simulations corroborate these findings and provide insights into the mechanism governing CO oxidation on IrO₂(110), predicting that bridging O-vacancies play an important role in mediating the reaction kinetics.

2. Experimental details

The experiments were performed in an UHV chamber with a base pressure of 2.0×10^{-10} Torr, the details of which have been reported previously [27–30]. Briefly, the chamber is equipped with an inductively-coupled RF plasma source (Oxford Scientific Instruments) for generating atomic oxygen beams, an electron-beam metal evaporator (McAllister Technical Services) for the deposition of metallic Ir, an ion source (SPECS), a low energy electron diffraction (LEED) optics (SPECS), and a shielded quadrupole mass spectrometer (Hiden HAL 511/3F PIC) for temperature programmed desorption (TPD) and reaction spectroscopy (TPRS). The chamber is also equipped with a dual Mg/Al anode x-ray source and a hemispherical analyzer (SPECS) for x-ray photoelectron spectroscopy (XPS).

2.1. IrO₂(110) film growth and surface structure

The circular Ru(0001) single crystal (9 mm × 1 mm) was attached to two 0.38 mm tungsten wires mounted onto an LN₂ cooled sample holder. A type K thermocouple was spot welded onto the back side of the sample for temperature measurements. Resistive heating, controlled using a PID controller that varies the output of a programmable DC power supply, supports linear ramping from 85 to 1450 K and maintaining a desired temperature. The sample was cleaned by alternating rounds of Ar⁺ sputtering (2 keV) and annealing in an O₂ background of 5×10^{-7} Torr both at 1000 K, and finally annealing in UHV at 1450 K for several minutes. This procedure has been shown to produce a high level of surface crystallinity and cleanliness as supported by LEED and XPS.

A stoichiometrically-terminated IrO₂(110) film [s-IrO₂(110)] was prepared for the present study using a “template-assisted growth” method wherein IrO₂(110) is grown epitaxially on a s-RuO₂(110) surface in UHV [31–33]. Close lattice matching allows IrO₂(110) to grow

epitaxially on the RuO₂(110) surface; an IrO₂(110) layer on RuO₂(110) is estimated to experience strain of only +1.6 % and -0.3 % along the [001] and $\bar{1}$ [10] directions, respectively [31–34]. In the first step of the growth, a s-RuO₂(110) thin film of about 4.2 nm thickness was generated by exposing clean Ru(0001) to an O-atom beam at 750 K. The film thickness of 4.2 nm is equivalent to about 13 “layers” of RuO₂(110), where one layer is equal to the spacing of 0.317 nm between Ru-containing planes along the [110] crystallographic direction of rutile RuO₂. In the next step, metallic Ir (Goodfellow, 99.9 %) was deposited from an e-beam evaporator onto the s-RuO₂(110) surface at 300 K followed by post-oxidation at 700 K using an O-atom beam to generate about 0.3 layers of IrO₂(110) domains on the surface. Thereafter, metallic Ir was deposited in an O₂ background (Airgas, 99.999 %) of 3×10^{-7} Torr at 700 K to produce a conformal IrO₂(110) thin film. In the final step, the IrO₂(110) film was exposed to an O-atom beam at 700 K and then exposed to 10 L (Langmuir) of O₂ at 600 K to further clean the surface and fill bridging oxygen (O_{br}) vacancies that may have formed. This procedure produced a high-quality IrO₂(110) thin film with a stoichiometric-termination and consisting of 8 layers of IrO₂(110), corresponding to a thickness of ~2.5 nm. The thickness of the IrO₂(110) thin film was estimated by quantifying the attenuation of the XPS Ru 3d signal, as reported previously [5,31]. XPS also shows that IrO₂ and RuO₂ in the film structure undergo negligible mixing at the temperatures employed in our experiments (≤ 700 K).

Fig. 1 shows a ball and stick model of IrO₂(110) with an adsorbed CO_t molecule next to an O_t atom. The IrO₂(110) surface has a rectangular unit cell with bulk-terminated dimensions of $a = 3.16$ Å and $b = 6.36$ Å and is composed of alternating rows of coordinatively unsaturated Ir (Ir_{cus}) and bridging O atoms (O_{br}) along the [001] direction. Both the Ir_{cus} and O_{br} atoms have single coordination vacancies due to a decrease in bond coordination relative to bulk IrO₂; the Ir_{cus} atoms have fivefold coordination, whereas bulk Ir atoms have sixfold coordination, and O_{br} atoms have twofold coordination, whereas bulk O-atoms have threefold coordination. Since Ir_{cus} atoms are the active adsorption sites, we define 1 ML as equal to the density of Ir_{cus} atoms on the IrO₂(110) surface. Thus, the initial coverages of Ir_{cus} and O_{br} atoms are each equal to 1 ML on the ideal s-IrO₂(110) surface. Additionally, the dissociative chemisorption of O₂ on the Ir_{cus} rows is facile and produces O-atoms (O_t) that adsorb directly on-top of Ir_{cus} atoms and co-exist with O_{br} atoms (Fig. 1) [4,22]. CO also adsorbs preferentially on top of Ir_{cus} atoms of s- and O_t-rich IrO₂(110) surfaces and can potentially react either with a neighboring O_{br} or O_t atom [4,7,23].

2.2. CO adsorption and oxidation

We investigated the oxidation of CO (BOC, 99.99 %) on the s-IrO₂(110) surface using TPRS after CO adsorption at 85 K. After CO exposure, TPRS spectra were collected by positioning the surface in front of a shielded mass spectrometer at a distance of ~5 mm and heating at a constant rate of 1 K/s up to 650 K. We note that a small amount of H₂ (< 0.1 ML) from the vacuum background adsorbs on the IrO₂(110) surface during cooling to 85 K prior to CO adsorption and desorbs as H₂O during TPRS experiments. After each TPRS experiment, the sample was exposed to ~20 L of O₂ while heating between 300 and 600 K to remove residual hydrogen and eliminate oxygen-vacancies that are created during TPRS. Prior work shows that this procedure generates O_t atoms that promote complete H₂O desorption below 600 K, allowing us to repeatedly generate a clean IrO₂(110) surface [4,5,21,22]. Reproducibility in our TPRS results provides evidence that IrO₂(110) films with nominally the same surface structure and composition can be repeatedly generated.

We also investigated CO oxidation on O_t-rich IrO₂(110) surfaces with different, initial on-top oxygen coverages. Surfaces of varying coverages of O_t atoms were prepared by first generating a saturated layer of O_t atoms by dosing 5 L of O₂ at 300 K, followed by flash heating to different temperatures between 300 K and 625 K at a rate of 1 K/s to controllably

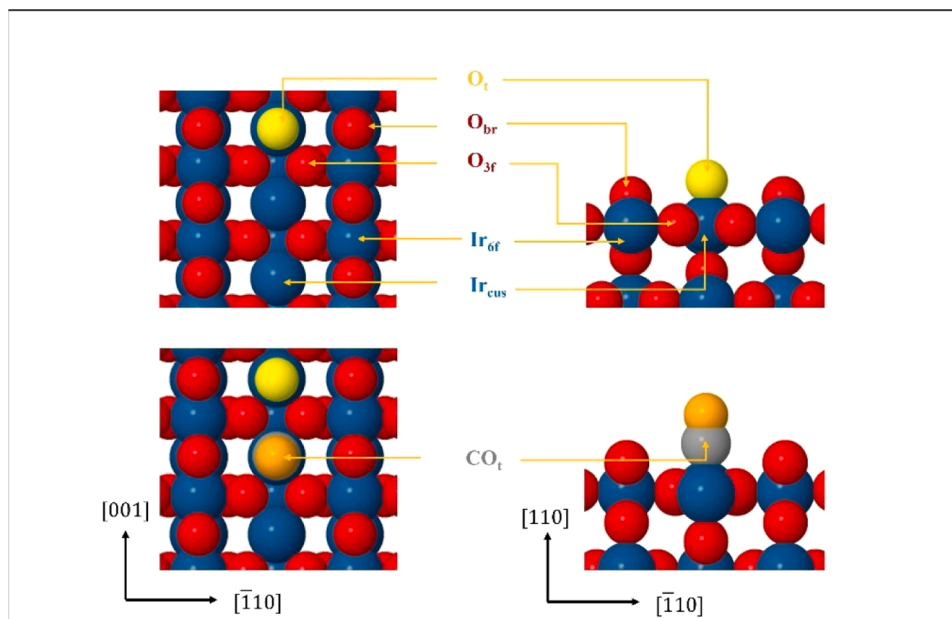


Fig. 1. Model representation of the top and side views of the $\text{IrO}_2(110)$ structure. Top: O_t - $\text{IrO}_2(110)$ surface with O_t , O_{3f} , O_{br} , Ir_{6f} and Ir_{cus} atoms indicated, where Ir_{6f} and O_{3f} correspond to six-fold and three-fold coordinated atoms, respectively. Bottom: CO_t adsorbed on O_t - $\text{IrO}_2(110)$ surface. Red, blue, and gray atoms represent O, Ir and C atoms, respectively. O_t atoms are colored yellow, and the O-atom of CO is colored orange.

desorb a known amount of oxygen from the O_t layer. The sample was then immediately cooled to 85 K for the adsorption of CO. The coverages of O_t atoms obtained as a function of flashing temperature were determined from separate temperature-programmed desorption (TPD) experiments. Prior studies establish that O_t atoms desorb completely from $\text{IrO}_2(110)$ upon heating to 625 K, whereas O_{br} atoms desorb negligibly below 625 K [1,6,35], which is analogous to the behavior of O_t and O_{br} atoms on $\text{RuO}_2(110)$ [12,13]. Coverages and desorption yields were quantified from the TPRS spectra using previously established procedures (Section S1).

3. Computational details

The projector augmented wave (PAW) pseudopotentials [36] within the Vienna ab initio simulation package (VASP) [37,38] were employed in all calculations. The exchange-correlation effects were treated using the Perdew-Burke-Ernzerhof (PBE) functional [39], and the plane wave expansion was limited to a cutoff energy of 400 eV. The $\text{IrO}_2(110)$ surface was modeled using a slab consisting of four atomic layers, providing a total thickness of approximately 12 Å, with a vacuum region of 25 Å to eliminate artificial periodic interactions along the z-axis. The lattice

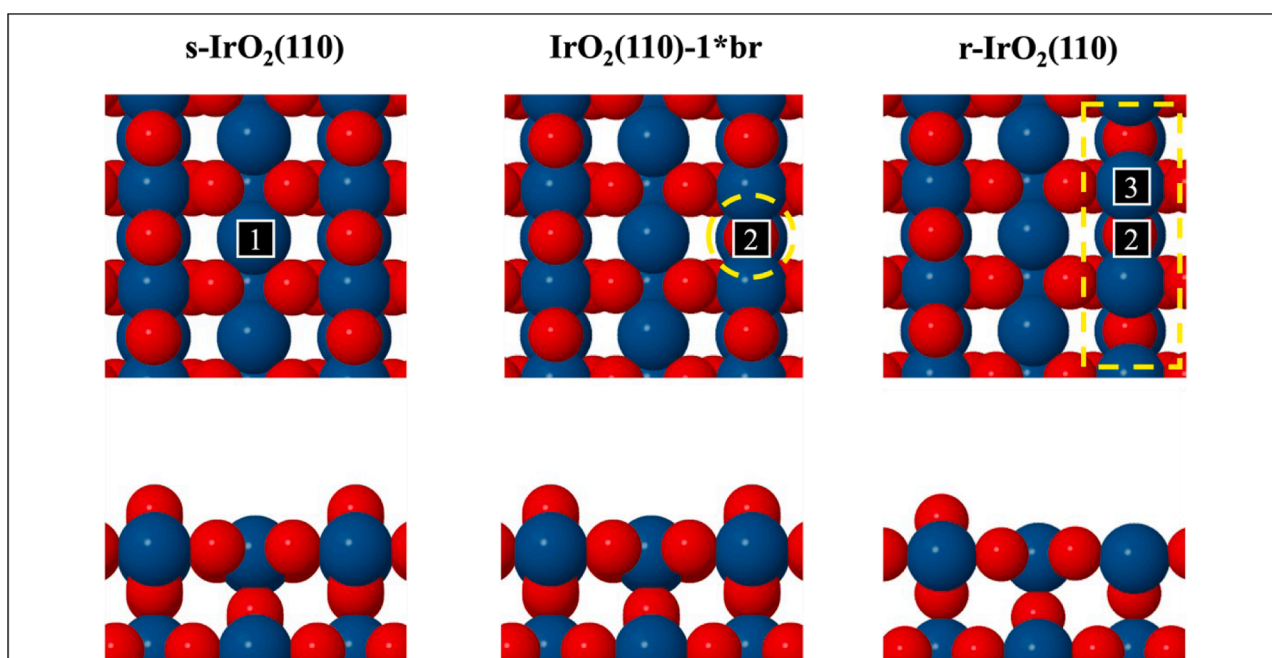


Fig. 2. Initial surface configurations of stoichiometric $\text{IrO}_2(110)$ ($\text{s-IrO}_2(110)$), $\text{IrO}_2(110)$ with a single oxygen vacancy ($\text{IrO}_2(110)-1^*\text{br}$), and partially-reduced $\text{IrO}_2(110)$ with an entire row of oxygen vacancies ($\text{r-IrO}_2(110)$). Vacancies are depicted with yellow dashed lines. Adsorption sites are depicted with black boxes with white lines, indicating (1) top site, (2) bridge symmetric site (br,sym), and (3) bridge atop site (br,atop).

constants ($a = 4.54 \text{ \AA}$ and $c = 3.19 \text{ \AA}$) from PBE calculations for bulk IrO_2 were used to set the lateral dimensions of the slab. During optimization, the bottom two layers were fixed, whereas the atoms in the remaining layers were allowed to relax until the forces were below $0.03 \text{ eV/\AA}$. A 4×2 unit-cell was implemented along with a $2 \times 2 \times 1$ Monkhorst-Pack k-point grid for Brillouin zone sampling.

In this study, we introduce several initial surface configurations to investigate CO adsorption on Ir_{cus} atoms as well as O_{br} -vacancies, designated as $^*\text{br}$ sites (Fig. 2). The first configuration that we modeled is the stoichiometric $\text{IrO}_2(110)$ surface ($s\text{-IrO}_2(110)$) with no vacancies. The second configuration involves $\text{IrO}_2(110)$ with a single oxygen vacancy (1^*br), where one O_{br} atom is removed. The third configuration represents a partially-reduced $\text{IrO}_2(110)$ surface ($r\text{-IrO}_2(110)$) with one entire row of O_{br} vacancies within the unit cell, simulating a highly reduced surface state. CO adsorption was examined on three types of sites: on top of an Ir_{cus} atom (CO_{t}), on one oxygen vacancy on the bridge row ($\text{CO}_{\text{br,sym}}$), and on top of Ir atoms on the reduced bridging row ($\text{CO}_{\text{br,atop}}$). On the $\text{IrO}_2(110)$ surface, an O_{br} atom is bonded in the bridge position above two underlying Ir atoms so removing one O_{br} atom creates an unfilled bridging site ($^*\text{br}$) relative to these subsurface Ir atoms as well as a coordination vacancy on each of the Ir atoms. Removing two neighboring O_{br} atoms produces two adjacent $^*\text{br}$ sites, where the Ir atom that separates these $^*\text{br}$ sites has two coordination vacancies.

The desorption energy (E_d) for molecules (CO , CO_2) from the surface was calculated using the formula:

$$E_d = (E_{\text{molecule}} + E_{\text{surf}}) - E_{\text{molecule/surf}}$$

where $E_{\text{molecule/surf}}$ is the total energy of the system with the adsorbed molecule, E_{surf} is the energy of the bare surface, and E_{molecule} is the energy of the molecule in the gas phase. A larger positive value for E_d indicates greater stability of the adsorbed molecule, and thus a higher desorption barrier. The reaction barriers (transition state barriers) on the $\text{IrO}_2(110)$ surface were calculated using the climbing nudged elastic band (cNEB) method [40]. Binding energies of CO in different configurations were also computed using the hybrid HSE06 functional in single-point calculations [41]. These calculations were motivated by prior studies showing that CO binding on the $\text{PdO}(101)$ surface is overestimated by calculations using the PBE functional, and that calculations with the HSE06 functional predict lower CO binding energies in better agreement with experimental estimates [42–44]. Section S2 in the SI presents test calculations of the impact of allowing relaxation in the HSE06 calculations.

4. Results

4.1. TPRS of CO oxidation on $s\text{-IrO}_2(110)$

Adsorbed CO on the $s\text{-IrO}_2(110)$ film desorbs or oxidizes during TPRS, and the branching toward CO_2 production decreases as the initial CO coverage is increased. At the lowest CO coverage studied (0.19 ML), CO oxidation produces a broad CO_2 TPRS feature with maxima at about 345 , 470 and 545 K , while unreacted CO desorbs in a smaller peak at 555 K that overlaps with the final CO_2 TPRS peak (Fig. 3a). As the initial CO coverage is increased, the CO and CO_2 TPRS peaks near 550 K intensify sharply, whereas the CO_2 TPRS features below $450\text{--}500 \text{ K}$ remain saturated at a CO_2 yield of ~ 0.05 to 0.10 ML (Section S3). At the saturation CO coverage (0.78 ML), the CO TPRS trace exhibits a single peak at 545 K , and the CO_2 TPRS trace exhibits a dominant peak at 545 K as well as a broad leading edge that extends to about 300 K (Fig. 3b). The CO TPRS peak downshifted slightly with increasing CO coverage, suggesting that CO molecules interact only weakly with one another on the $s\text{-IrO}_2(110)$ surface. About 38% of the adsorbed CO oxidized to CO_2 during TPRS when the CO layer was initially saturated, demonstrating that CO desorption becomes increasingly favored over CO_2 production as the initial CO coverage is increased.

The main CO and CO_2 TPRS peaks at about 545 K are attributed to the desorption and oxidation of CO on the majority $s\text{-IrO}_2(110)$ phase, while CO_2 production below about 500 K is attributed mainly to CO oxidation on metallic Ir domains that are assumed to be present in small quantities within the oxide film. The emergence of CO and CO_2 in single TPRS peaks near 545 K is generally consistent with DFT calculations of CO desorption and oxidation on the $s\text{-IrO}_2(110)$ surface (see below) [4,7,11,21,45,46]. The observed CO peak temperature of 545 K is also close to, though slightly lower than, that reported for CO desorption from Ir-atoms substitutionally-doped into $\text{Fe}_3\text{O}_4(001)$, for which the surface Ir atoms adopt an octahedral geometry that resembles that of the Ir_{cus} atoms of $\text{IrO}_2(110)$ [47].

Although we do not have evidence from spectroscopy or microscopy that metallic Ir domains are present within the $s\text{-IrO}_2(110)$ film, prior STM studies show that small metallic pits or islands develop on $\text{RuO}_2(110)$ and $\text{IrO}_2(110)$ after mild reduction and heating to $> 600 \text{ K}$ [33,48,49]. Also, we performed separate measurements which show that CO oxidation occurs efficiently on O-covered $\text{Ir}(100)$, with CO_2 desorbing in multiple TPRS features between ~ 200 and 500 K after CO was adsorbed to saturation at $\sim 85 \text{ K}$ (Section S4). It should be noted that a CO_2 yield of 0.05 to 0.10 ML would correspond to about 0.02 to 0.04 CO_2 molecules per surface Ir atom of a metallic Ir domain because the Ir_{cus} site density of $\text{IrO}_2(110)$ is about three times lower than the atomic

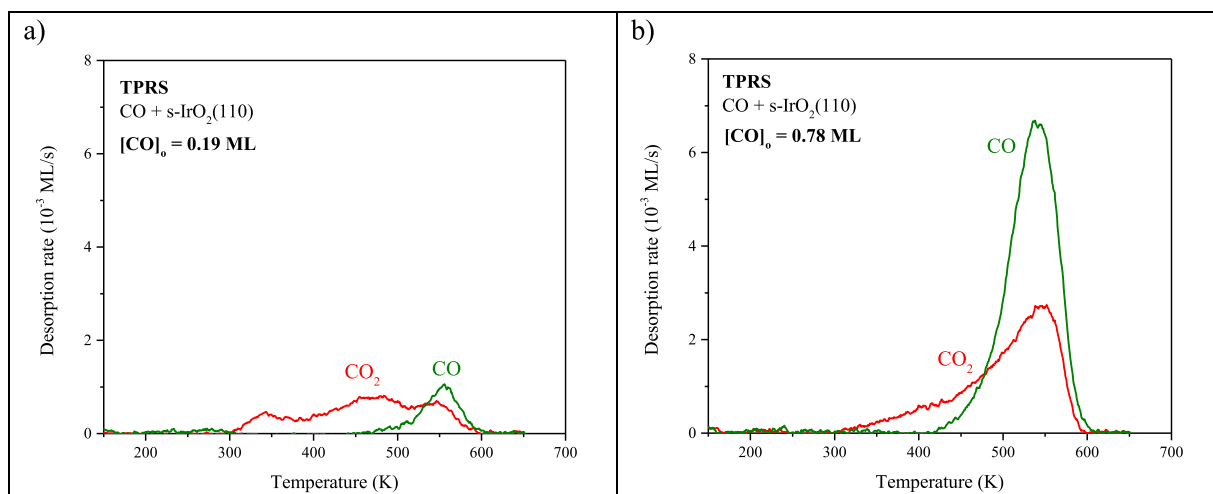


Fig. 3. CO (green) and CO_2 (red) TPRS traces obtained after adsorbing CO on $s\text{-IrO}_2(110)$ at 85 K to generate initial CO coverages of a) 0.19 ML and b) 0.78 ML .

density of a low-index Ir metal surface. Thus, metallic Ir could cover as little as 2 % of the surface area and still generate CO_2 in the quantities observed below ~ 500 K during TPRS from the s- $\text{IrO}_2(110)$ film.

4.2. TPRS product yields from CO oxidation on s- $\text{IrO}_2(110)$

Quantitative analysis shows that CO adsorbs in high coverages on the s- $\text{IrO}_2(110)$ film. As seen in Fig. 4a, the total CO coverage generated at 85 K increased sharply with increasing CO exposure, until saturating at a value of 0.78 ML. Given that CO preferentially adsorbs on top of the Ir_{cus} atoms [4,23], the high saturation coverage indicates that CO occupies nearly all of the Ir_{cus} sites when the CO layer is saturated. The fact that the CO TPRS peak downshifts only slightly as the CO coverage is increased (Fig. 3, Section S3) further suggests that neighboring CO molecules interact weakly along the Ir_{cus} rows. Notably, we have previously reported that chemisorbed N_2 adopts a similar, linear bonding geometry as CO on the Ir_{cus} sites of $\text{IrO}_2(110)$, and that N_2 also achieves a saturation coverage of about 0.8 ML [31]. Prior to the CO exposures, the adsorption of H_2 from the vacuum background likely occupies a fraction of the Ir_{cus} sites, and prevents CO from reaching the ideal saturation coverage of 1 ML.

Our results show that CO desorption becomes increasingly favored over CO_2 production as the CO coverage is increased. At the lowest CO coverage studied (0.19 ML), the CO_2 production yield was nearly twice as large as the CO desorption yield (Fig. 4a). However, the CO desorption yield increased sharply as the CO coverage was increased, and surpassed the CO_2 production yield at CO coverages above ~ 0.4 ML (Fig. 4a). The fractional yield of CO_2 generated from s- $\text{IrO}_2(110)$ during TPRS is 72 % at the lowest CO coverage, and decreased to 38 % once the initial CO coverage was increased to saturation (Fig. 4b), at which point about 0.48 ML of CO desorbed and 0.29 ML oxidized to CO_2 during TPRS. The decrease in the CO_2 production yield can be attributed generally to the corresponding decrease in the amount of reactive O_{br} atoms available to each adsorbed CO molecule as the CO coverage is increased. Interestingly, the fractional CO_2 yield at the CO saturation coverage is lower (~ 38 % vs. 70 %) than that reported for the stoichiometrically-terminated $\text{RuO}_2(110)$ and $\text{PdO}(101)$ surfaces [24, 25,43,44], suggesting that the s- $\text{IrO}_2(110)$ surface is intrinsically less active toward CO oxidation than these analogous reactive metal-oxides. This difference may originate from the energetics for CO oxidation on these oxide surfaces or other factors associated with the reaction mechanism.

4.3. TPRS of CO oxidation on O_t -rich $\text{IrO}_2(110)$

Pre-adsorbed O_t atoms promote the oxidation of CO on $\text{IrO}_2(110)$ during TPRS. Fig. 5a and 5b show CO and CO_2 TPRS traces acquired after adsorbing CO to saturation at 85 K on $\text{IrO}_2(110)$ surfaces initially with zero vs. 0.28 ML of O_t atoms. As discussed above, CO and CO_2 evolve from the s- $\text{IrO}_2(110)$ surface in peaks at 545 K, and only 38 % of the adsorbed CO oxidizes to CO_2 by reaction with the oxide surface (Fig. 5a). Pre-adsorbed O_t atoms promote CO oxidation over desorption and enable CO_2 production at lower temperature compared with reaction on s- $\text{IrO}_2(110)$. At an initial O_t coverage of 0.28 ML, CO oxidation generates a dominant CO_2 TPRS peak at 370 K, while CO_2 and CO desorption near 545 K is strongly diminished (Fig. 5b). The CO_2 peak at 370 K is attributed to CO oxidation by O_t atoms. The CO saturation coverage was lower on the $\text{IrO}_2(110)$ surface with 0.28 vs. 0 ML of O_t atoms (0.41 vs. 0.78 ML), due to site blocking by O_t atoms (see below). However, at comparable initial CO coverages (~ 0.4 ML), the fractional CO_2 yield obtained from CO oxidation was still significantly higher on the O_t -rich compared with s- $\text{IrO}_2(110)$ surfaces (94 % vs. 50 %), demonstrating that O_t atoms facilitate the oxidation of CO adsorbed on $\text{IrO}_2(110)$.

4.4. TPRS product yields from CO oxidation on O_t -rich $\text{IrO}_2(110)$

Pre-adsorbed O_t atoms suppress CO adsorption on $\text{IrO}_2(110)$ but significantly promote CO oxidation over desorption. The initial CO coverage generated on $\text{IrO}_2(110)$ decreases linearly as the pre-coverage of O_t atoms is increased to saturation at about 0.5 ML (Fig. 6a); the corresponding TPRS traces are discussed in detail below. This behavior strongly suggests that CO preferentially binds on Ir_{cus} sites of $\text{IrO}_2(110)$ and that occupation of these sites by O_t atoms prevents CO adsorption. Pre-adsorbed O_t atoms also inhibit the adsorption of CH_4 , C_2H_6 and N_2 on O_t -rich $\text{IrO}_2(110)$ due to Ir_{cus} site blocking [4,5,21,22,31]. Although the initial CO coverage decreased, the CO_2 yield increased to a maximum value of about 0.4 ML as the O_t coverage was increased to 0.25–0.30 ML, while the CO desorption yield concurrently decreased to nearly zero (Fig. 6a). The CO_2 yield then decreased with increasing O_t coverage above ~ 0.3 ML. The strong enhancement in CO oxidation over desorption is further illustrated by the sharp increase in the fractional CO_2 yield from about 38 % to 100 % as the initial O_t coverage is increased to 0.4 ML (Fig. 6b). Recombinative desorption of O_t atoms was only observed for initial O_t coverages above ~ 0.4 ML, when the fractional CO_2 yield reached 100 % and the initial CO coverage fell below the initial O_t coverage (Section S5). This behavior demonstrates that O_t

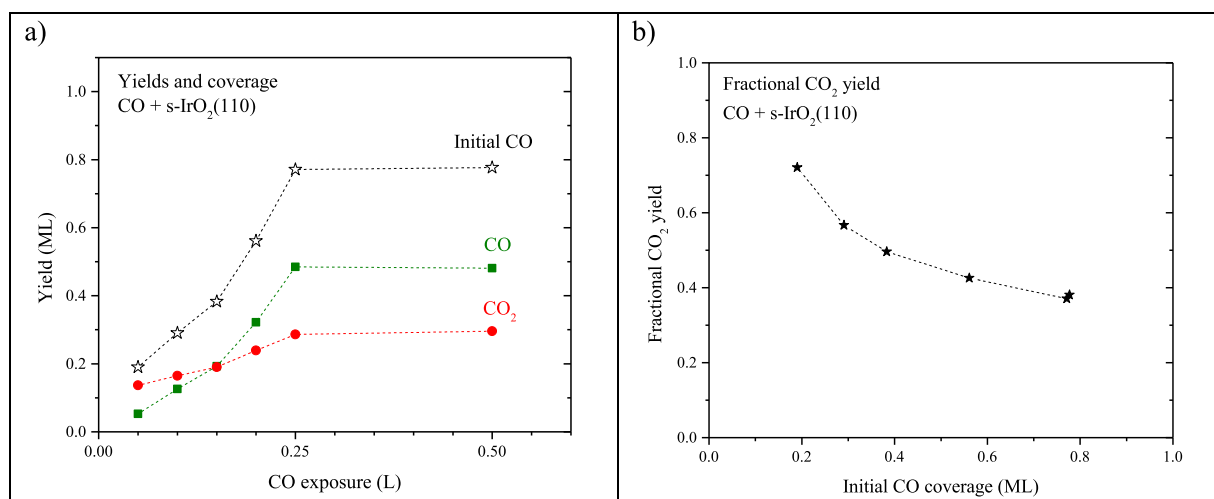


Fig. 4. a) Initial CO coverage (black) and the yields of desorbed CO (green) and CO_2 (red) shown as a function of CO exposure (L) for TPRS experiments of CO adsorbed on s- $\text{IrO}_2(110)$ at 85 K. b) The fractional yield of CO_2 produced from CO oxidation on s- $\text{IrO}_2(110)$ as a function of the initial CO coverage.

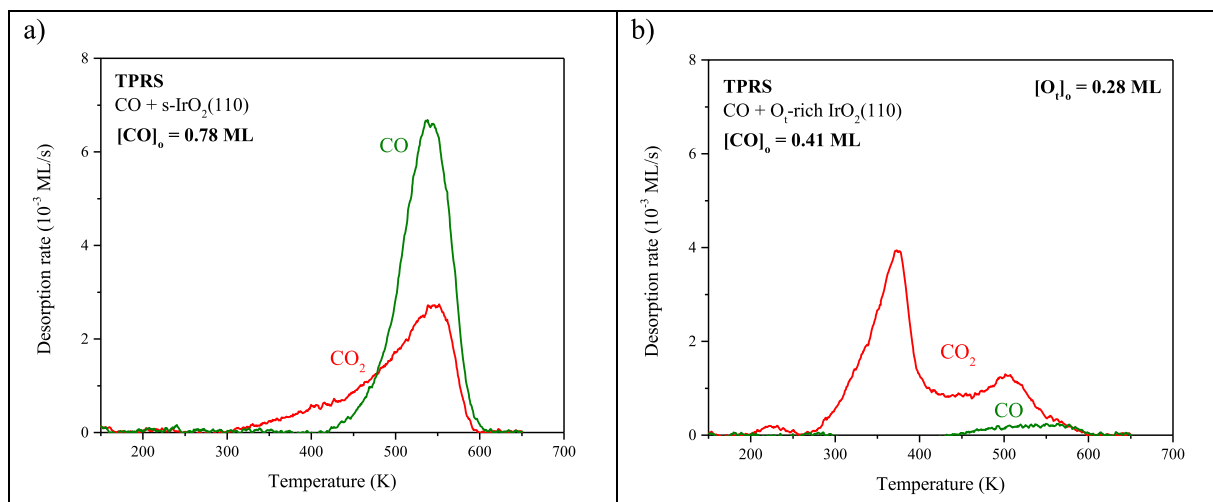


Fig. 5. CO (green) and CO₂ (red) TPRS traces obtained after adsorbing CO to saturation at 85 K on IrO₂(110) surfaces with initial O_i coverages of a) 0 ML and b) 0.28 ML.

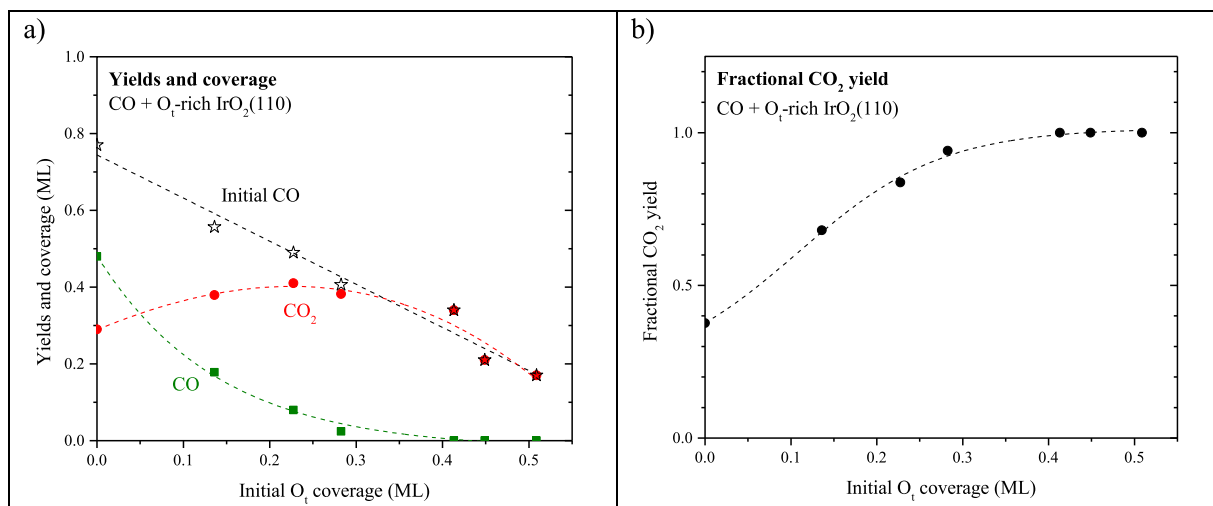


Fig. 6. a) Initial CO coverage (black) and the yields of desorbed CO (green) and CO₂ (red) shown as a function of the initial O_i coverage for TPRS experiments of CO adsorbed to saturation on O_i-rich IrO₂(110) at 85 K. b) The fractional yield of CO₂ produced from CO oxidation on O_i-rich IrO₂(110) as a function of the initial O_i coverage.

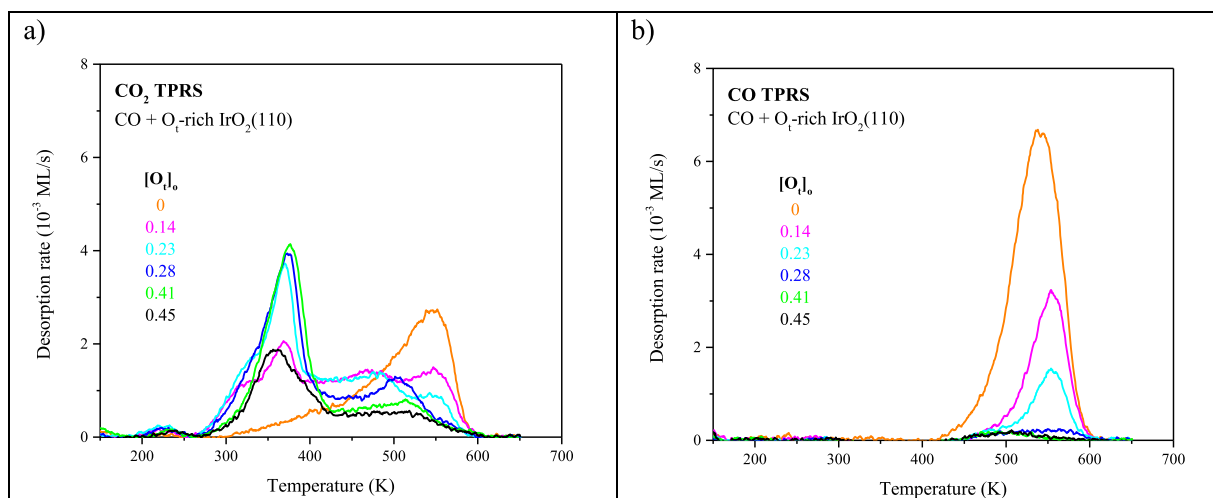


Fig. 7. a) CO₂ and b) CO TPRS traces obtained after adsorbing CO to saturation at 85 K on IrO₂(110) surfaces with different initial O_i coverages.

atoms are readily subsumed by CO oxidation. In part, the increase in CO oxidation activity arises because a larger amount of reactive oxygen (O_t and O_{br}) is available at a given CO coverage on O_t -rich $IrO_2(110)$.

4.5. TPRS as a function of initial O_t coverage from CO on O_t -rich $IrO_2(110)$

TPRS suggests that a lower energy pathway governs CO oxidation by O_t vs. O_{br} atoms on $IrO_2(110)$, and consequently that adsorbed CO molecules preferentially react with O_t atoms during TPRS on O_t -rich $IrO_2(110)$. As seen in Fig. 7a, the CO_2 TPRS peak near 545 K diminishes, while a sharp CO_2 peak at 370 K concurrently intensifies as the O_t coverage is initially increased. The CO_2 TPRS peak at 370 K then becomes less intense as the O_t coverage is increased above ~ 0.4 ML and the initial CO coverage further decreases. Small features in the CO_2 TPRS traces, e.g., near 325 and 475 K, likely originate from CO oxidation on minority quantities of metallic Ir, as discussed above for the s- $IrO_2(110)$ surface. The CO TPRS peak near 545 K also diminishes sharply as the O_t coverage is increased (Fig. 7b), demonstrating the preference for CO oxidation when O_t atoms are available. Overall, this behavior provides evidence that the oxidation of CO adsorbed on $IrO_2(110)$ involves a significantly lower barrier when reaction abstracts an O_t vs. an O_{br} atom from the surface [4,7,23]. A consequence is that CO_2 desorbs at lower temperature when CO reacts with an O_t rather than an O_{br} atom (370 vs. 545 K) during TPRS. The lower reaction barrier also causes CO oxidation to become kinetically favored over desorption, as evidenced by the fact that the fraction of CO that oxidizes to CO_2 increases with the O_t coverage (Fig. 6b).

4.6. DFT calculations of CO binding on $IrO_2(110)$

DFT predicts that CO preferentially bonds on top of Ir_{cus} atoms of the stoichiometric and O_t -rich $IrO_2(110)$ surfaces, and that the atop CO species (CO_t) is slightly stabilized vs. destabilized by neighboring O_t vs. CO species on the Ir_{cus} row. In its minimum energy configuration, CO adopts a linear geometry on top of an Ir_{cus} atom of s- $IrO_2(110)$ and achieves a binding energy of 223 kJ/mol according to DFT-PBE calculations (Fig. 8a). Positioning the CO_t species next to 1 vs. 2 O_t atoms increases its binding energy to 231 and 236 kJ/mol (Fig. 8b), respectively, demonstrating that O_t atoms mildly stabilize the atop CO species on $IrO_2(110)$. In contrast, neighboring CO_t molecules slightly destabilize one another on $IrO_2(110)$ as the differential binding energy decreases from 223 to 211 kJ/mol as the coverage of CO_t on s- $IrO_2(110)$ is increased to 1 ML (Section S6). The coverage dependent changes in the CO binding energy on stoichiometric and O_t -rich $IrO_2(110)$ would cause a change of about 35 K in the CO TPRS peak temperature, which is larger than our experimental observations (~ 15 K).

Our DFT calculations show that CO can also achieve stable binding on vacant sites ($*_{br}$) within the O_{br} row of partially-reduced $IrO_2(110)$ surfaces, suggesting the possibility that CO can populate multiple states on $IrO_2(110)$ during CO oxidation. Analogous binding configurations have been reported for CO adsorbed on partially-reduced $RuO_2(110)$, and shown to play a role in the kinetics of CO oxidation on this surface [25,50,51]. We investigated the adsorption of CO on an $IrO_2(110)$ surface with a single $*_{br}$ site [$IrO_2(110)-1*_{br}$] and a partially-reduced $IrO_2(110)$ surface [r- $IrO_2(110)$] in which all O_{br} atoms of one row of the (4×2) slab model were removed (Fig. 2). According to DFT-PBE, CO adopts a symmetric bridging geometry on the $*_{br}$ site of the $IrO_2(110)-1*_{br}$ surface, and achieves a binding energy of 212 kJ/mol;

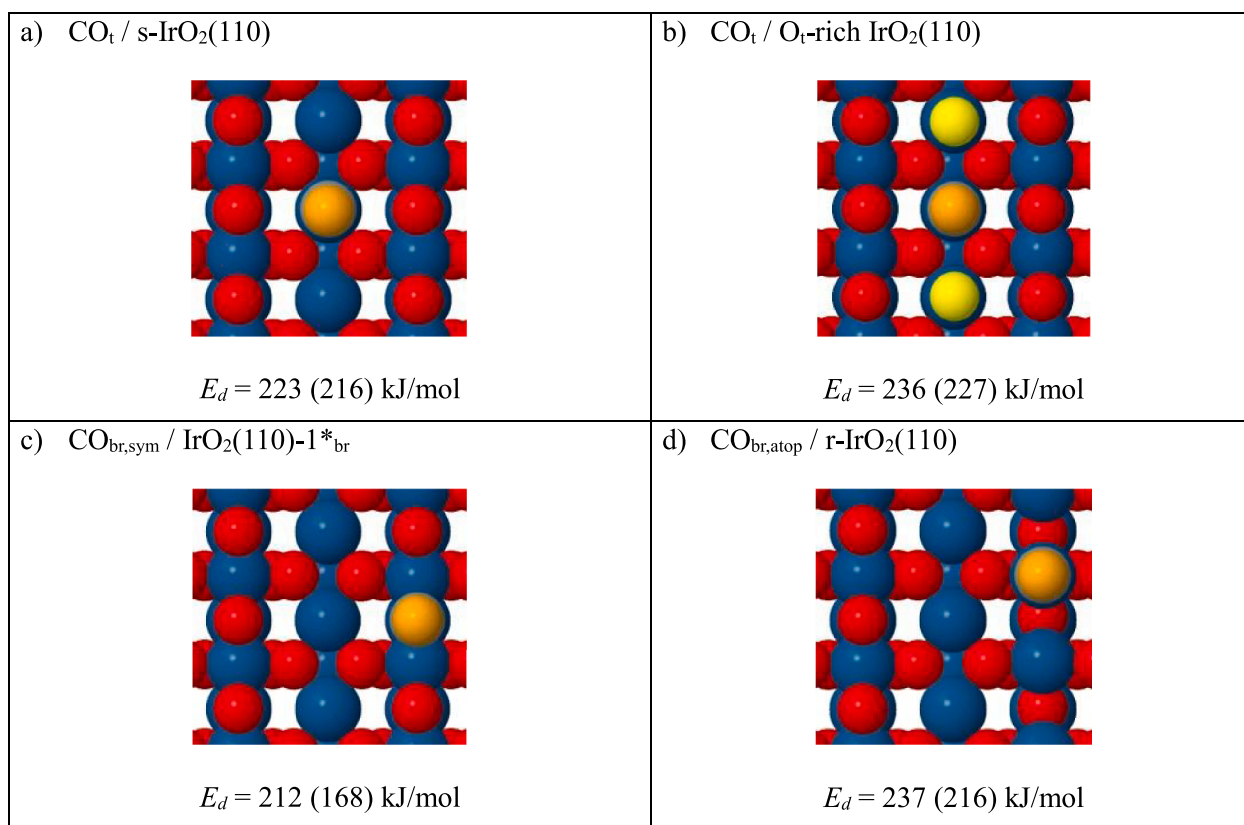


Fig. 8. Images of CO adsorbed on $IrO_2(110)$ surfaces and the CO binding energies, i.e., desorption barriers, determined from DFT-PBE calculations. Binding energies determined from single-point HSE06 calculations are shown in parentheses. a) CO_t on s- $IrO_2(110)$, b) CO_t with 2 adjacent O_t atoms on $IrO_2(110)$, c) $CO_{br,sym}$ species on $IrO_2(110)-1*_{br}$ and d) $CO_{br,atop}$ species on r- $IrO_2(110)$. Red, blue, and gray atoms represent O, Ir and C atoms, respectively. O_t atoms are colored yellow, and the O-atom of CO is colored orange.

this CO binding configuration is designated as the $\text{CO}_{\text{br,sym}}$ species (Fig. 8c). DFT-PBE predicts a similar binding energy of 206 kJ/mol for a single $\text{CO}_{\text{br,sym}}$ species on the r- $\text{IrO}_2(110)$ surface, demonstrating that neighboring O_{br} atoms only weakly affect the $\text{CO}_{\text{br,sym}}$ binding. Relative to the $^*\text{br}$ site, CO achieves a higher binding energy of 237 kJ/mol, according to DFT-PBE, by adsorbing on top of an undercoordinated Ir atom on the $^*\text{br}$ row of r- $\text{IrO}_2(110)$ to produce the so-called $\text{CO}_{\text{br,atop}}$ species (Fig. 8d). For CO adsorbed on the r- $\text{IrO}_2(110)$ surface, DFT predicts significant repulsion and in particular for $\text{CO}_{\text{br,atop}}$ the differential binding energy decreases from 237 to 61 kJ/mol as the coverage is increased to 1 ML (Section S6). However, such high coverages of $\text{CO}_{\text{br,atop}}$ would require extended rows of O_{br} vacancies and this likely is a minor contribution to the CO TPRS spectra. Overall, the similarity in the binding energies of the various CO configurations suggests that CO may occupy multiple types of sites on partially-reduced $\text{IrO}_2(110)$ surfaces.

Binding energies of CO in different configurations were also computed using the hybrid HSE06 functional in single-point calculations (Section S2). These calculations were motivated by prior studies showing that CO binding on the $\text{PdO}(101)$ surface is overestimated by calculations using the PBE functional, and that calculations with the HSE06 functional predict lower CO binding energies in better agreement with experimental estimates [42–44]. Analysis of the TPRS spectra also suggest that DFT-PBE overestimates the CO binding energies on $\text{IrO}_2(110)$. Assuming desorption pre-factors (A) of $\log A = 15$ to 17, we estimate that a CO binding energy between about 170 and 190 kJ/mol would produce a TPRS peak temperature of 545 K. A single-point HSE06 calculation predicts only a slight lowering of the binding energies of the atop CO species relative to the PBE values; the HSE06 and PBE binding energies are 216 vs. 223 kJ/mol for the CO_{t} species and 216 vs. 237 kJ/mol for the $\text{CO}_{\text{br,atop}}$ species. In contrast, the HSE06 calculation predicts a greater destabilization of the bridging $\text{CO}_{\text{br,sym}}$ species relative to the PBE result, as the $\text{CO}_{\text{br,sym}}$ binding energy decreased from 212 to 168 kJ/mol (PBE vs. HSE06). This behavior is consistent with our earlier work comparing HSE06 and PBE calculations of the binding of bridging vs. atop CO on $\text{PdO}(101)$ [42–44]. A key finding is that the binding energy of the $\text{CO}_{\text{br,sym}}$ species determined using a single-point HSE06 calculation lies near the range estimated from our experimental TPRS results.

4.7. DFT calculations of CO oxidation on stoichiometric and partially-reduced $\text{IrO}_2(110)$

DFT predicts that CO oxidation is favored over desorption on $\text{IrO}_2(110)$ surfaces. In the absence of $^*\text{br}$ sites, a CO_{t} species can either desorb or oxidize to CO_2 by reaction with a neighboring O_{br} atom. In the CO oxidation pathway, the CO_{t} species reacts with an O_{br} atom to generate a surface CO_2 species ($\text{CO}_{2,\text{br}}$), which can subsequently desorb to produce a gaseous CO_2 molecule and a $^*\text{br}$ site (Fig. 9a). The surface CO_2 species generated by this reaction is designated as $\text{CO}_{2,\text{br}}$ because an O-atom of this species is located close to the initial position of the O_{br} atom involved in reaction. Our DFT calculations predict that reaction between CO_{t} and O_{br} to generate $\text{CO}_{2,\text{br}}$ has a barrier of 95 kJ/mol while the reverse reaction has a negligible barrier. Since the barrier for desorption of the $\text{CO}_{2,\text{br}}$ species is 68 kJ/mol, the reaction between CO_{t} and O_{br} atom to produce gaseous CO_2 is effectively a single step process with a barrier of 163 kJ/mol. An apparent reaction barrier (E_{app}) may be defined as the difference between the intrinsic reaction and desorption barriers, $E_{\text{app}} = E_{\text{r}} - E_{\text{d}}$, and provides insights into the kinetic branching probability for reaction. For CO_{t} oxidation by an O_{br} atom, the apparent reaction barrier is highly negative at $E_{\text{app}} = -60$ kJ/mol, and thus suggests that CO_{t} oxidation is strongly favored over desorption on s- $\text{IrO}_2(110)$. When computed using the HSE06 binding energy of the CO_{t} species, the apparent barrier remains low ($E_{\text{app}} = -53$ kJ/mol) and a large preference for CO_{t} oxidation is still predicted.

We also identified pathways for CO_{t} hopping to vacant sites on the bridging row, and the desorption or oxidation of the resulting CO_{br} species. DFT-PBE predicts that conversion of a CO_{t} species to a $\text{CO}_{\text{br,sym}}$ species via hopping to a neighboring $^*\text{br}$ site has a barrier of 183 kJ/mol and is endothermic by 11 kJ/mol on the $\text{IrO}_2(110)$ -1 $^*\text{br}$ surface (Fig. 9b). Subsequent reaction of the $\text{CO}_{\text{br,sym}}$ species with an O_{br} atom involves a barrier of 148 kJ/mol and is endothermic by 82 kJ/mol. This reaction produces a surface CO_2 species ($\text{CO}_{2,\text{br,sym}}$) that is aligned parallel to the bridging row, but can transform rapidly to the more stable $\text{CO}_{2,\text{br}}$ species by overcoming a barrier of only 40–50 kJ/mol (Section S7). For clarity, Fig. 9b omits the $\text{CO}_{2,\text{br,sym}}$ species and its conversion to $\text{CO}_{2,\text{br}}$. The $\text{CO}_{2,\text{br}}$ species must then overcome a 64 kJ/mol barrier to desorb, generating a gaseous CO_2 molecule and 2 $^*\text{br}$ sites. For this pathway, the effective barrier for $\text{CO}_{\text{br,sym}}$ oxidation to gaseous CO_2 is 148 kJ/mol, and the apparent reaction barrier is $E_{\text{app}} = -62$ kJ/mol when computed using the PBE value of the $\text{CO}_{\text{br,sym}}$ binding energy. Similar to the CO_{t} species, the

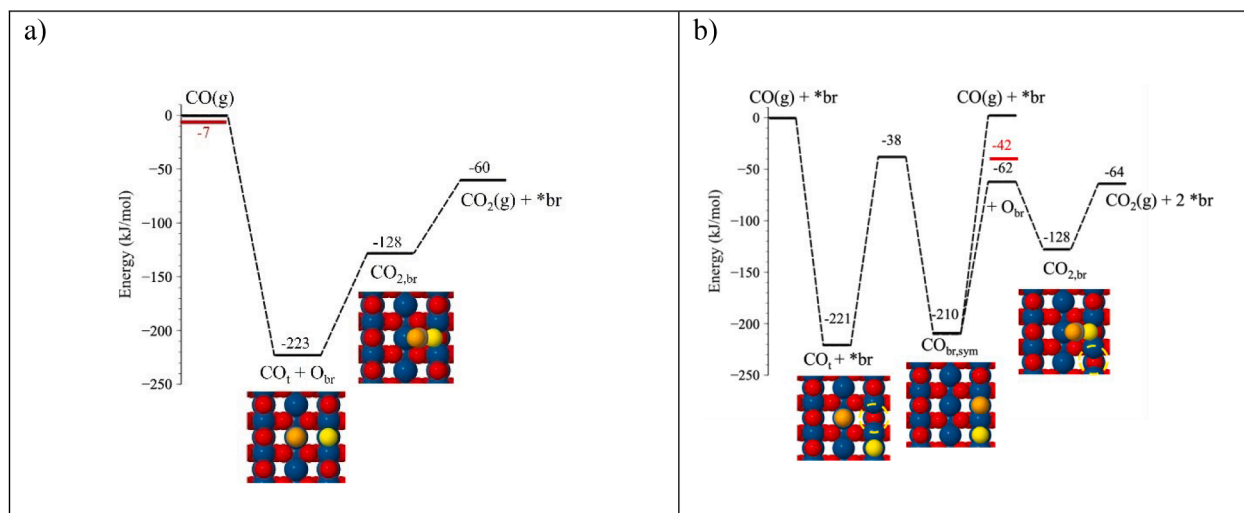


Fig. 9. Energy diagrams for a) oxidation of a CO_{t} molecule by an O_{br} atom of s- $\text{IrO}_2(110)$, and b) conversion of CO_{t} to $\text{CO}_{\text{br,sym}}$ and oxidation of the $\text{CO}_{\text{br,sym}}$ species by a neighboring O_{br} atom. Energy levels shown in red represent the final state of $\text{CO}(\text{g})$ relative to the a) CO_{t} and b) $\text{CO}_{\text{br,sym}}$ initial states determined from single-point HSE06 calculations. O_{br} atoms involved in the reactions are depicted in yellow, the O-atom of CO is depicted in orange and O_{br} vacancies ($^*\text{br}$) are indicated by dashed yellow circles.

highly negative value of E_{app} suggests that oxidation of $\text{CO}_{\text{br,sym}}$ is strongly favored over desorption. However, E_{app} for $\text{CO}_{\text{br,sym}}$ oxidation increases to -20 kJ/mol when computed using the HSE06 binding energy (168 kJ/mol), which suggests a more equitable kinetic competition between $\text{CO}_{\text{br,sym}}$ desorption and oxidation. Below, we show that the experimental branching between CO desorption and oxidation observed during TPRS from s- $\text{IrO}_2(110)$ can be reproduced by microkinetic simulations only when the CO binding energies are lowered from their PBE values.

According to DFT, barriers for CO diffusion are high on $\text{IrO}_2(110)$ so the adsorbed CO species are relatively immobile. DFT-PBE predicts that the barrier for a CO_t molecule to hop to a neighboring Ir_{cus} atom is 200 kJ/mol, which is nearly 90 % of the CO_t binding energy. A potential consequence is that, in the absence of O_t atoms, the CO_t species can only oxidize directly on $\text{IrO}_2(110)$ if a neighboring O_{br} atom is available since CO_t is kinetically hindered from diffusing along the Ir_{cus} row to locate another O_{br} reaction partner. Further, the barriers for CO_t hopping to the bridging row and converting to a $\text{CO}_{\text{br,sym}}$ or $\text{CO}_{\text{br,atop}}$ species are also large at about 182 kJ/mol, but smaller than the CO_t hopping barrier along the Ir_{cus} row. An implication is that conversion of CO_t to a CO_{br} species is kinetically favored when both neighboring O_{br} atoms are removed since CO_t desorption and site hopping must overcome larger barriers. The large barriers for CO diffusion generally suggest that the kinetics for CO oxidation on $\text{IrO}_2(110)$ may be strongly dependent on the local configurations of adsorbed CO species.

4.8. CO oxidation on O_t -rich $\text{IrO}_2(110)$

DFT suggests that CO oxidation is more efficient on O_t -rich than stoichiometric $\text{IrO}_2(110)$ because the barrier is significantly lower for CO_t oxidation by an O_t vs. an O_{br} atom. The reaction of CO_t with an O_t atom generates an $\text{CO}_{2,t}$ species, and involves a barrier of 62 kJ/mol and has an exothermicity of -20 kJ/mol (Fig. 10). Notably, the low barrier (82 kJ/mol) for $\text{CO}_{2,t}$ dissociation into CO_t and O_t suggests that s- $\text{IrO}_2(110)$ may be effective in activating CO_2 at elevated pressure for catalytic applications. The apparent barrier for CO_t oxidation by an O_t atom is highly negative at $E_{app} = -169$ kJ/mol, indicating that 100 % of CO_t molecules adsorbed next to O_t atoms will oxidize to CO_2 rather than desorb. Also, the intrinsic reaction barrier is about 100 kJ/mol lower for CO_t reaction with an O_t vs. an O_{br} atom. Thus, the DFT calculations

confirm that the large downshift in the CO_2 peak temperature as well as the strong enhancement in CO oxidation activity of O_t -rich vs. stoichiometric $\text{IrO}_2(110)$ originates from a significantly lower barrier for CO_t oxidation by O_t vs. O_{br} atoms.

4.9. TPRS simulations: CO oxidation on s- $\text{IrO}_2(110)$

A microkinetic model was developed from the DFT results to simulate TPRS traces during CO oxidation on stoichiometric and O_t -rich $\text{IrO}_2(110)$. A description of the reaction steps, rate equations and kinetic parameters that were employed in the model is given in Section S8. The model sets an initial coverage of CO_t species and O_t atoms, and includes steps for CO_t desorption and oxidation by an O_{br} atom. Steps are also included for the conversion of CO_t to $\text{CO}_{\text{br,sym}}$ or $\text{CO}_{\text{br,atop}}$ as well as the desorption and oxidation of $\text{CO}_{\text{br,sym}}$ and the desorption of $\text{CO}_{\text{br,atop}}$. Steps associated with CO_t oxidation by an O_t atom are also included in the model. For the CO binding energies, we ran simulations using values determined from PBE and single-point HSE06 calculations (Section S2), while all other energy barriers were set equal to the PBE values, with one exception as discussed below. No coverage dependence of CO desorption was included. As noted earlier, CO adsorbed in O_{br} vacancies do show coverage dependence but in our microkinetic simulations the coverage of CO_{br} is never sufficiently high for this to play an appreciable role. Prefactors for the desorption and reaction steps were set to physically reasonable values, and adjusted slightly to improve the agreement with our experimental results.

Our TPRS simulations provide evidence that the CO binding energies determined from DFT-PBE calculations are too high, and that lowering them is necessary to improve agreement with the experimental observations. Specifically, when using the CO desorption barriers from PBE, our simulations predict negligible CO desorption and that more than 99 % of the adsorbed CO oxidizes to CO_2 on s- $\text{IrO}_2(110)$ during TPRS for all initial CO coverages (Section S9). In contrast, our experimental data shows that the fractional CO_2 yield is only 38 % when the initial CO layer is saturated on s- $\text{IrO}_2(110)$, and that CO and CO_2 desorb concurrently in peaks at about 545 K. The overly high CO oxidation activity predicted by the simulations can be attributed partly to the fact that PBE predicts highly negative values of the apparent reaction barriers.

Lowering the barriers for both $\text{CO}_{\text{br,sym}}$ desorption and the conversion of CO_t to $\text{CO}_{\text{br,sym}}$ significantly improves the agreement between the simulated and experimental TPRS traces. We specifically set the CO desorption barriers equal to the CO binding energies determined from single-point HSE06 calculations. This change causes the desorption barrier of the $\text{CO}_{\text{br,sym}}$ species to decrease significantly (168 vs. 212 kJ/mol), thus increasing the apparent reaction barrier ($E_{app} = -20$ kJ/mol) and shifting the kinetic branching of the $\text{CO}_{\text{br,sym}}$ species toward desorption. However, lowering only the CO desorption barriers did not enhance CO desorption in the simulations. The reason is that the PBE barrier for CO_t oxidation is lower than that for CO_t conversion to $\text{CO}_{\text{br,sym}}$ (163 vs. 183 kJ/mol) so CO_2 production remained dominant and the $\text{CO}_{\text{br,sym}}$ state populated negligibly in the simulations.

The desorption of CO became appreciable during the simulations when the barrier for CO_t conversion to $\text{CO}_{\text{br,sym}}$ was lowered, and the $\text{CO}_{\text{br,sym}}$ desorption barrier was set equal to the HSE value. Specifically, we obtain good quantitative agreement with the experimental CO and CO_2 TPRS yields when the CO_t to $\text{CO}_{\text{br,sym}}$ conversion barrier used in the simulations was lowered to 76 % of its PBE value, which corresponds to a decrease from 183 to 139 kJ/mol. For example, for an initial CO coverage of 0.6 ML, the simulations predict that CO_2 desorbs in a peak at 542 K and CO desorbs in a sharper peak at 551 K (Fig. 11a), and that the fractional CO_2 yield is equal to 36 %. TPRS traces simulated for several initial CO coverages are shown in Section S9. Overall, the use of lower barriers for both $\text{CO}_{\text{br,sym}}$ formation and desorption increases the overall CO desorption rates predicted by the simulations, and thereby provides a more accurate representation of the kinetic competition between CO desorption vs. oxidation on s- $\text{IrO}_2(110)$ with CO and CO_2 desorbing

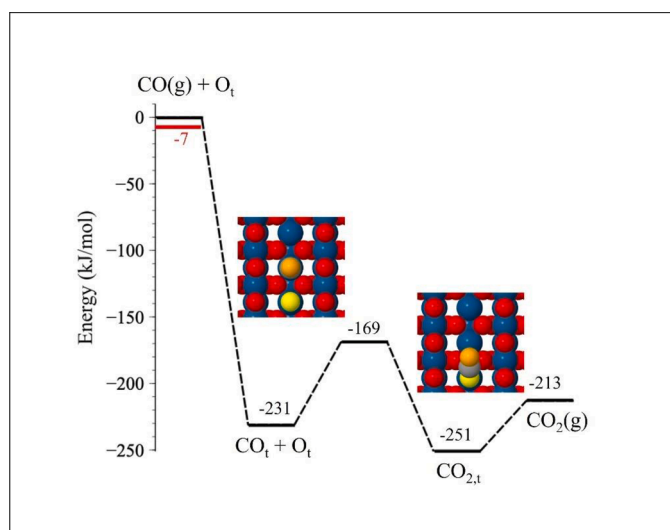


Fig. 10. Energy diagram for oxidation of a CO_t molecule by an O_t atom on $\text{IrO}_2(110)$. The energy level shown in red represents the final state of $\text{CO}(\text{g})$ relative to the CO_t initial state determined from a single-point HSE06 calculation. The O_t atom and the O-atom of CO are depicted in yellow and orange, respectively.

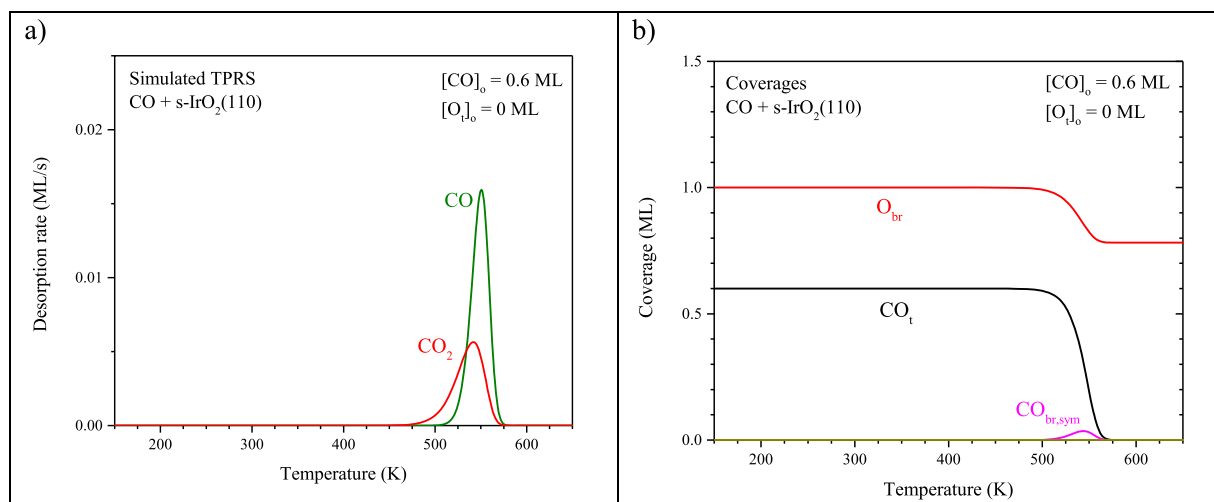


Fig. 11. a) Simulated CO and CO₂ TPRS traces for CO oxidation on s-IrO₂(110) for an initial CO coverage of 0.6 ML. The microkinetic model uses CO desorption barriers determined from single-point HSE06 calculations, and a barrier for CO_t to CO_{br,sym} conversion that is 76 % of the value determined from PBE. All other barriers were determined with PBE calculations. b) Coverages of O_{br}, CO_t and CO_{br,sym} species during the TPRS simulation.

nearly concurrently during TPRS.

Our simulations predict that conversion of CO_t to CO_{br,sym} plays a central role in determining the branching between CO desorption vs. oxidation on s-IrO₂(110). Initially, reaction between CO_t and O_{br} atoms is kinetically favored in the simulations since the reaction barrier is less than the desorption barrier for the CO_t species (163 vs. 216 kJ/mol). As a result, CO₂ desorption initiates at lower temperature than CO desorption (Fig. 11a), causing the CO_t and O_{br} coverages to decrease sharply with temperature (Fig. 11b). As *_{br} sites are created by CO₂ production, CO_t conversion to CO_{br,sym} begins to occur at comparable rates as CO_t oxidation and the CO_{br,sym} coverage increases as a result. The CO_{br,sym} coverage then decreases as desorption and oxidation consume the CO_{br,sym} species (Fig. 11b). The CO_{br,sym} coverages remain relatively low during the simulations (<~ 0.1 ML), demonstrating that the CO_{br,sym} species promptly desorb or oxidize at the temperatures at which they form.

According to the simulations, CO₂ evolution from s-IrO₂(110) arises predominantly from the direct reaction between CO_t and O_{br} species, whereas CO desorption occurs entirely from the CO_{br,sym} state. This behavior originates from the differences in the apparent reaction barriers of the CO_t and CO_{br,sym} species ($E_{app} = -53$ vs. -20 kJ/mol), and thus the branching probability for reaction. For a typical apparent pre-factor of reaction ($A_r/A_d = 10^{-2}$) [44,52,53], the branching probability for reaction remains nearly unity for the CO_t species up to at least 600 K, but decreases sharply for the CO_{br,sym} species as the temperature is increased above 300 K. The decrease in O_{br} coverage as CO₂ is produced during TPRS also lowers the rate of CO_{br,sym} oxidation, thus further increasing the branching probability for CO_{br,sym} desorption.

It should be stressed that our simulations utilized energy barriers that were computed from DFT, and that only the barrier for CO_t to CO_{br,sym} conversion was adjusted to improve agreement with the experimental TPRS data. We show that the microkinetic simulations capture key characteristics of the TPRS traces for CO oxidation on s-IrO₂(110) when the model uses CO desorption barriers determined from HSE06 single-point calculations and the barrier for CO_t to CO_{br,sym} conversion is lowered from its PBE value; all other barriers were set to the PBE values. A similar approach was previously shown to provide a more accurate representation of the binding and oxidation of CO on the PdO(101) surface [42–44]. This prior work lends confidence to the TPRS simulations presented here for CO oxidation on IrO₂(110), but additional work is needed to further explore the reaction mechanism.

4.10. TPRS simulations: CO oxidation on O_t-rich IrO₂(110)

The simulations predict that CO_t oxidation by O_t atoms generates CO₂ at low temperature and increases the fractional yield of CO₂ at a given initial CO coverage. Fig. 12a and 12b show TPRS traces and species coverages determined from the kinetic simulations for initial CO and O_t coverages of 0.6 ML and 0.2 ML, respectively. Reaction between CO_t and O_t produces a CO₂ TPRS peak at 243 K, and completely consumes the O_t atoms for these initial conditions. The remaining CO_t then oxidizes by reaction with O_{br} or converts to the CO_{br,sym} state and desorbs, resulting in the CO₂ and CO peaks at higher temperature. As expected, the fractional CO₂ yield obtained for a given initial CO coverage is higher when O_t atoms are present on the surface. For [CO]₀ = 0.6 ML, the fractional CO₂ yield is 60 % vs. 36 % for initial O_t coverages of 0.2 ML vs. 0 ML (Figs. 11 and 12), and increases to 100 % when [O_t]₀ ≥ [CO]₀, in good agreement with the behavior observed experimentally. The CO₂ peak temperature resulting from CO_t oxidation by O_t is significantly lower in the simulations compared with experiment (243 vs. 370 K), suggesting that DFT-PBE underestimates the barrier for this reaction. Despite this difference, the simulations support the conclusion that O_t atoms promote CO oxidation on IrO₂(110) by providing a lower energy pathway for reaction.

5. Discussion

Our results demonstrate that IrO₂(110) is less active for CO oxidation during TPRS than the related RuO₂(110) and PdO(101) surfaces. Both CO and the CO₂ reaction product desorb during TPRS at much higher temperatures from s-IrO₂(110) compared with RuO₂(110) and PdO(101). For s-IrO₂(110), CO desorption and oxidation produce CO and CO₂ TPRS peaks near 545 K, whereas these reactions generate CO and CO₂ peaks near 300 K in TPRS measurements from both RuO₂(110) and PdO(101) [13,25,42–44]. In accord with the BEP relation, these results suggest that the stronger interaction of CO with IrO₂(110) increases the barriers for both CO desorption and oxidation, and thus causes CO and CO₂ to desorb at higher temperatures from IrO₂(110) relative to RuO₂(110) and PdO(101). Future studies to clarify the electronic factors that determine CO binding strengths on these oxide surfaces would be worthwhile to pursue. Although reaction with O_t rather than O_{br} atoms provides a significantly lower energy pathway for CO oxidation on IrO₂(110), CO oxidation by O_t atoms still produces CO₂ at higher temperatures (~370 K) on IrO₂(110) compared with O_t-rich RuO₂(110) (~300 K) [24,25]. Consistent with the TPRS behavior, a recent study

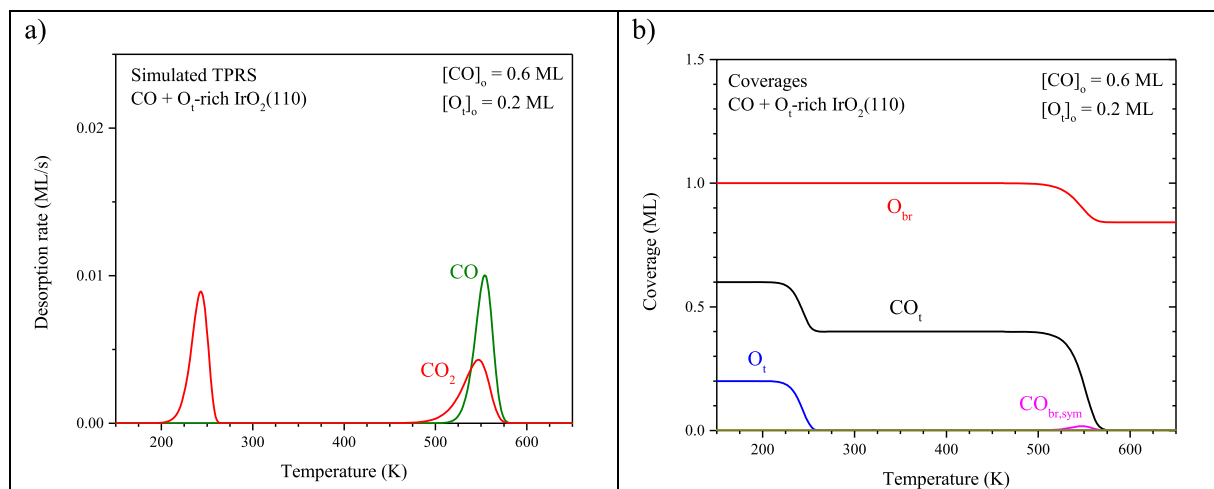


Fig. 12. a) Simulated CO and CO_2 TPRS traces for CO oxidation on O_t -rich $\text{IrO}_2(110)$ for an initial CO coverage of 0.6 ML and an initial O_t coverage of 0.2 ML. The microkinetic model uses CO desorption barriers determined from single-point HSE06 calculations, and a barrier for CO_t to $\text{CO}_{\text{br,sym}}$ conversion that is 76 % of the value determined from PBE. All other barriers were determined with PBE calculations. b) Coverages of O_{br} , O_t , CO_t and $\text{CO}_{\text{br,sym}}$ species during the TPRS simulation.

with TiO_2 -supported oxide powders demonstrates that rutile IrO_2 is less active than RuO_2 for catalyzing CO oxidation; the apparent activation energy is reported to be about 65 kJ/mol higher for CO oxidation on pure IrO_2 compared with RuO_2 at stoichiometric feed conditions [20]. The present work suggests that stronger bonding with CO is mainly responsible for lowering the CO oxidation activity of $\text{IrO}_2(110)$ relative to $\text{RuO}_2(110)$ and $\text{PdO}(101)$.

Our results also suggest that similar rate processes influence CO oxidation on $\text{IrO}_2(110)$ compared with $\text{RuO}_2(110)$ and $\text{PdO}(101)$. Prior studies show that CO molecules fill O-vacancies during CO oxidation on $\text{RuO}_2(110)$ and that the resulting CO_{br} species must overcome larger barriers than the CO_t species to desorb or oxidize [13,25,54,55]. For $\text{RuO}_2(110)$, the barriers for CO_t vs. CO_{br} are sufficiently different that the desorption and oxidation of these species produce well-separated CO and CO_2 TPRS features near ~300 vs. 500 K, respectively [12,13,25]. Analogous behavior has been reported for CO oxidation on $\text{PdO}(101)$ in that CO adsorbed next to an O-atom vs. an O-vacancy produces distinct CO and CO_2 TPRS features at similar temperatures as reported for $\text{RuO}_2(110)$ [42,43].

Our DFT results and kinetic simulations predict that CO_{br} species are also generated during CO oxidation on $\text{IrO}_2(110)$, and play an important role in mediating the kinetics. However, according to our kinetic simulations, the CO_{br} species have a different influence on the CO oxidation kinetics on $\text{IrO}_2(110)$ compared with $\text{RuO}_2(110)$ and $\text{PdO}(101)$. On these latter surfaces, the CO species bind more strongly (~40–70 kJ/mol) near/on O-vacancies relative to the pristine surfaces, and their probability for oxidizing to CO_2 is near or greater than 50 % since the apparent reaction barriers are highly negative [25,42,43,50,51]. In contrast, DFT calculations using both the PBE and HSE06 functionals predict that $\text{CO}_{\text{br,sym}}$ binds more weakly by ~10 to 45 kJ/mol than CO_t on $\text{IrO}_2(110)$. As discussed above, the $\text{CO}_{\text{br,sym}}$ binding energy determined from HSE06 calculations is only 20 kJ/mol greater than its oxidation barrier, and, as a result, nearly all of the $\text{CO}_{\text{br,sym}}$ species desorb during TPRS simulations. Furthermore, since the barriers for CO_t oxidation and $\text{CO}_{\text{br,sym}}$ desorption used in the simulations have similar values of 163 and 168 kJ/mol, respectively, $\text{CO}_{\text{br,sym}}$ desorbs over a similar temperature range as CO_t is oxidized to CO_2 , and produces nearly overlapping CO and CO_2 TPRS peaks.

Key implications of our study are that the CO oxidation mechanism on $\text{IrO}_2(110)$, $\text{RuO}_2(110)$ and $\text{PdO}(101)$ involves similar reaction steps, but that relations among the energy barriers of these steps are different on $\text{IrO}_2(110)$ compared with the other oxides and that distinct CO oxidation kinetics manifest as a result. The weaker binding of the CO_{br} ,

sym vs. CO_t species on $\text{IrO}_2(110)$ has a decisive influence on the simulated kinetics. Our work also highlights the need to benchmark the accuracy of various exchange-correlation functionals for predicting binding energies and reaction barriers of adsorbates on the $\text{IrO}_2(110)$ surface through comparison with experimental data. Indeed, measurements using surface vibrational spectroscopy have been essential for identifying CO in different local environments (e.g., CO_t vs. CO_{br}) on $\text{RuO}_2(110)$ and $\text{PdO}(101)$ during CO oxidation [42,43,54–57]. Similar experiments are needed to test the prediction that CO_{br} species form during CO oxidation on $\text{IrO}_2(110)$, and are the topic of future work by our groups.

6. Summary

We investigated the desorption and oxidation of CO on $\text{IrO}_2(110)$ surfaces using TPRS, DFT calculations and microkinetic simulations. Adsorbed CO on s- $\text{IrO}_2(110)$ produces CO and CO_2 TPRS peaks at about 545 K, and only about 38 % of the CO oxidizes when the CO layer is initially saturated. Pre-adsorbed O_t atoms enhance the CO oxidation activity of $\text{IrO}_2(110)$ by providing a low energy pathway for reaction. The oxidation of CO by O_t atoms produces a CO_2 TPRS peak at 370 K and the fractional yield of CO_2 generated by CO oxidation increases to 100 % once the initial O_t coverage becomes greater than the CO coverage. In agreement with the experimental results, DFT predicts that the barrier is about 100 kJ/mol lower for CO oxidation by an O_t vs. O_{br} atom on $\text{IrO}_2(110)$.

Our results suggest that DFT-PBE calculations overestimate the binding energies of CO on $\text{IrO}_2(110)$, and that the lower values determined from single-point HSE06 calculations are more accurate. This conclusion is based on our finding that TPRS simulations can accurately reproduce the branching between CO desorption and CO_2 production on s- $\text{IrO}_2(110)$, only when the model employs the CO binding energies determined from HSE06 calculations. The TPRS simulations predict that CO_2 production on s- $\text{IrO}_2(110)$ originates predominantly from the oxidation of CO_t species and that CO desorption occurs only from the $\text{CO}_{\text{br,sym}}$ state, which is populated during TPRS when CO_t molecules fill $^*_{\text{br}}$ sites generated by CO_2 formation. Our calculations suggest that future experiments should be directed toward identifying CO_t and CO_{br} species, and clarifying their roles in mediating CO oxidation on $\text{IrO}_2(110)$. This study provides new insights for understanding the mechanism of CO oxidation on $\text{IrO}_2(110)$, revealing similarities and key differences in the behavior of CO on other reactive oxide surfaces.

CRediT authorship contribution statement

Connor Pope: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jungwon Yun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Rishikishore Reddy:** Validation, Investigation, Data curation. **Jovenal Jamir:** Validation, Investigation, Data curation. **Minkyu Kim:** Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Aravind Asthagiri:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jason F. Weaver:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jason Weaver reports financial support and equipment, drugs, or supplies were provided by University of Florida. Jason Weaver reports financial support was provided by US Department of Energy Basic Energy Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.susc.2024.122619](https://doi.org/10.1016/j.susc.2024.122619).

References

- [1] Z. Liang, T. Li, M. Kim, A. Asthagiri, J.F. Weaver, *Science* 356 (2017) 299–303.
- [2] Y.X. Bian, M. Kim, T. Li, A. Asthagiri, J.F. Weaver, *J. Am. Chem. Soc.* 140 (2018) 2665–2672.
- [3] R. Martin, M. Kim, A. Franklin, Y. Bian, A. Asthagiri, J.F. Weaver, *Phys. Chem. Chem. Phys.* 20 (2018) 29264–29273.
- [4] M. Kim, A. Franklin, R. Martin, F. Feng, T. Li, Z. Liang, A. Asthagiri, J. Weaver, *J. Phys. Chem. C* 123 (2019) 27603–27614.
- [5] C.J. Lee, S. Vashishtha, M. Shariff, F.R. Zou, J.J. Shi, R.J. Meyer, J.F. Weaver, *J. Phys.: Condens. Matter* 34 (2022).
- [6] Y.X. Bian, T. Li, J.F. Weaver, *J. Phys. D: Appl. Phys.* 52 (2019).
- [7] J. Jamir, M. Kim, C. Pope, A. Asthagiri, J.F. Weaver, *ACS Catal* 13 (2023) 14598–14613.
- [8] R. Martin, C.J. Lee, V. Mehar, M. Kim, A. Asthagiri, J.F. Weaver, *ACS Catal* 12 (2022) 2840–2853.
- [9] R. Martin, J. Jamir, M. Kim, C.J. Lee, V. Mehar, A. Asthagiri, J.F. Weaver, *J. Phys. Chem. C* 126 (2022) 15156–15166.
- [10] J. Jamir, C. Pope, S. Ramasubramanian, V. Mehar, J.J. Shi, J.F. Weaver, *ACS Catal* 14 (2024) 7062–7073.
- [11] M. Kim, A.D. Franklin, R. Martin, Y.X. Bian, J.F. Weaver, A. Asthagiri, *J. Catal.* 383 (2020) 181–192.
- [12] J.F. Weaver, *Chem. Rev.* 113 (2013) 4164–4215.
- [13] H. Over, *Chem. Rev.* 112 (2012) 3356–3426.
- [14] H. Over, Y.D. Kim, A.P. Seitsonen, S. Wendt, E. Lundgren, M. Schmid, P. Varga, A. Morgante, G. Ertl, *Science* 287 (2000) 1474–1476.
- [15] H. Over, M. Muhler, *Prog. Surf. Sci.* 72 (2003) 3–17.
- [16] F. Gao, D.W. Goodman, *Langmuir* 26 (2010) 16540–16551.
- [17] F. Gao, S.M. McClure, Y. Cai, K.K. Gath, Y. Wang, M.S. Chen, Q.L. Guo, D. W. Goodman, *Surf. Sci.* 603 (2009) 65–70.
- [18] M. Shipilin, et al., *J. Phys. Chem. C* 119 (2015) 15469–15476.
- [19] J. Gustafson, et al., *ACS Catal* 8 (2018) 4438–4445.
- [20] O. Khalid, T. Weber, G. Drazic, I. Djerdj, H. Over, *J. Phys. Chem. C* 124 (2020) 18670–18683.
- [21] J. Jamir, J. Yun, C. Pope, M. Kim, A. Asthagiri, J.F. Weaver, *J. Phys. Chem. C* 128 (2024) 111–122.
- [22] T. Li, M. Kim, Z. Liang, A. Asthagiri, J. Weaver, *Catal., Struct. React.* 4 (2018) 1–13.
- [23] C.H. Yeh, B.C. Ji, S. Nachimuthu, J.C. Jiang, *Appl. Surf. Sci.* (2021) 539.
- [24] S. Wendt, M. Knapp, H. Over, *J. Am. Chem. Soc.* 126 (2004) 1537–1541.
- [25] S. Wendt, A.P. Seitsonen, Y.D. Kim, M. Knapp, H. Idriss, H. Over, *Surf. Sci.* 505 (2002) 137–152.
- [26] M. Rieger, J. Rogal, K. Reuter, *Phys. Rev. Lett.* 100 (2008) 016105.
- [27] A.L. Gerrard, J.J. Chen, J.F. Weaver, *J. Phys. Chem. B* 109 (2005) 8017–8028.
- [28] R. Rai, T. Li, Z. Liang, M. Kim, A. Asthagiri, J.F. Weaver, *Surf. Sci.* 652 (2016) 213–221.
- [29] H.H. Kan, R.B. Shumera, J.F. Weaver, *Surf. Sci.* 602 (2008) 1337–1346.
- [30] W. Cartas, R. Rai, A. Sathe, A. Schaefer, J.F. Weaver, *J. Phys. Chem. C* 118 (2014) 20916–20926.
- [31] R. Martin, M. Kim, C.J. Lee, M.S. Shariff, F. Feng, R.J. Meyer, A. Asthagiri, J. F. Weaver, *J. Chem. Phys.* 152 (2020).
- [32] M.J.S. Abb, B. Herd, H. Over, *J. Phys. Chem. C* 122 (2018) 14725–14732.
- [33] M.J.S. Abb, et al., *J. Phys. Chem. C* 124 (2020) 15324–15336.
- [34] F.G. Sen, A. Kinaci, B. Narayanan, S.K. Gray, M.J. Davis, S.K.R. S. Sankaranarayanan, M.K.Y. Chan, *J. Mater. Chem. A* 3 (2015) 18970–18982.
- [35] T. Li, M. Kim, Z. Liang, A. Asthagiri, J.F. Weaver, *Top. Catal.* 61 (2018) 397–411.
- [36] P.E. Blochl, *Phys. Rev. B* 50 (1994) 17953–17979.
- [37] G. Kresse, *J. Non-Cryst. Solids* 193 (1995) 222–229.
- [38] G. Kresse, J. Hafner, *J. Non-Cryst. Solids* 156 (1993) 956–960.
- [39] J.P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* 77 (1996) 3865.
- [40] G. Henkelman, B.P. Uberuaga, H. Jonsson, *J. Chem. Phys.* 113 (2000) 9901–9904.
- [41] A.V. Krukau, O.A. Vydrov, A.F. Izmaylov, G.E. Scuseria, *J. Chem. Phys.* 125 (2006).
- [42] J.F. Weaver, F. Zhang, L. Pan, T. Li, A. Asthagiri, *Acc. Chem. Res.* 48 (2015) 1515–1523.
- [43] F. Zhang, L. Pan, T. Li, J. Diulus, A. Asthagiri, J.F. Weaver, *J. Phys. Chem. C* 118 (2014) 28647–28661.
- [44] V. Mehar, et al., *ACS Catal* 8 (2018) 8553–8567.
- [45] R. Martin, M. Kim, A. Asthagiri, J.F. Weaver, *ACS Catal* 11 (2021) 4682–4703.
- [46] S.D. Senanayake, J.A. Rodriguez, J.F. Weaver, *Acc. Chem. Res.* 53 (2020) 1488–1497.
- [47] J. Hulva, M. Meier, R. Bliem, Z. Jakub, F. Kraushofer, M. Schmid, U. Diebold, C. Franchini, G.S. Parkinson, *Science* 371 (2021) 375.
- [48] H. Over, A.P. Seitsonen, E. Lundgren, M. Schmid, P. Varga, *Surf. Sci.* 515 (2002) 143–156.
- [49] M. Knapp, D. Crihan, A.P. Seitsonen, E. Lundgren, A. Resta, J.N. Andersen, H. Over, *J. Phys. Chem. C* 111 (2007) 5363–5373.
- [50] A.P. Seitsonen, Y.D. Kim, M. Knapp, S. Wendt, H. Over, *Phys. Rev. B* 65 (2002) 035413.
- [51] C.Y. Fan, J. Wang, K. Jacobi, G. Ertl, *J. Chem. Phys.* 114 (2001) 10058–10062.
- [52] J.F. Weaver, A.F. Carlsson, R.J. Madix, *Surf. Sci. Rep.* 50 (2003) 107–199.
- [53] C.T. Campbell, Y.K. Sun, W.H. Weinberg, *Chem. Phys. Lett.* 179 (1991) 53–57.
- [54] A. Farkas, F. Hess, H. Over, *J. Phys. Chem. C* 116 (2012) 581–591.
- [55] A. Farkas, G.C. Mellau, H. Over, *J. Phys. Chem. C* 113 (2009) 14341–14355.
- [56] J.F. Weaver, J. Choi, V. Mehar, C.J. Wu, *ACS Catal* 7 (2017) 7319–7331.
- [57] J. Choi, L. Pan, V. Mehar, F. Zhang, A. Asthagiri, J.F. Weaver, *Surf. Sci.* 650 (2016) 203–209.